

Osmania University

REGULAR EXAMINATION ? 2026

Design of Reinforced Concrete Structures ? Regular Examination, 2026

Subject Code: CE402 Semester: 5 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Limit State Method)

1. Define Limit State Method. Explain its significance in Design of Reinforced Concrete Structures with a suitable example. [12 Marks]
2. How is Limit State Method implemented? Give an algorithm / procedure and explain. [8 Marks]
3. Compare and contrast the key concepts involved in Limit State Method. [12 Marks]

SECTION B (Unit 2: Flexure Design of Beams)

4. Explain Flexure Design of Beams with a neat diagram/sketch and a suitable example. [12 Marks]
5. Explain Flexure Design of Beams with a neat diagram/sketch and a suitable example. [12 Marks]
6. Discuss the engineering applications of Flexure Design of Beams. [12 Marks]

SECTION C (Unit 3: Shear and Torsion Design)

7. Solve a problem related to Shear and Torsion Design. Show all the steps clearly. [12 Marks]
8. Derive the expression related to Shear and Torsion Design and explain each term. [12 Marks]
9. Describe the working principle of Shear and Torsion Design with necessary equations. [8 Marks]

SECTION D (Unit 4: Slabs and Columns)

10. Derive the expression related to Slabs and Columns and explain each term. [12 Marks]
11. Compare and contrast the key concepts involved in Slabs and Columns. [12 Marks]
12. Write short notes on Slabs and Columns. [12 Marks]

SECTION E (Unit 5: Prestressed Concrete)

13. List the advantages and limitations of Prestressed Concrete. Explain any two in detail. [8 Marks]
14. Discuss the engineering applications of Prestressed Concrete. [8 Marks]
15. Discuss the engineering applications of Prestressed Concrete. [8 Marks]

*** End of Question Paper ***